Does a Popular Trading Signal Beat Buy-and-Hold?

Survivorship-Free Evidence on the Triple Supertrend from the Nifty 50, 2010–2026

Abhinav Gudipati*

October 2026

Abstract

The Triple Supertrend, a trend-following rule that trades only when three Average True Range bands agree, is widely used by Indian retail traders. We test it on an equal-weighted portfolio of the stocks that were members of the Nifty 50 on each day from January 2010 to October 2026, using National Stock Exchange (NSE) end-of-day files adjusted for splits, bonuses, rights issues, demergers and dividends with NSE's own corporate-action records. The panel covers 99.3% of index member-days, including companies later delisted. Parameters are chosen on 2010–2018 and tested unchanged on 2019–2026. Out of sample, the strategy returns 8.0% a year against 15.1% for buy-and-hold, with volatility of 8.5% against 17.4% and a maximum drawdown of −12.1% against −38.6%. The Sharpe ratios (0.96 and 0.90) are statistically indistinguishable. Back-filling today's constituents overstates buy-and-hold returns by 7–10 percentage points a year. Buying index options on the same signals loses money in simulation. The rule behaves as a risk-reduction overlay rather than a source of excess return.

Keywords: technical analysis, trend following, Supertrend, survivorship bias, Indian equities, Nifty 50
JEL classification: G11, G14, G17

*Affiliation. E-mail: gabhinavsiddharth@gmail.com This is a working paper; comments are welcome. This paper is research, not investment advice.

1 Introduction

Supertrend is one of the most widely used indicators among Indian retail traders. It is built into most Indian broking platforms, and a common variant stacks three Supertrends of different sensitivity, trading only when all three agree. Despite this popularity, we are not aware of a peer-reviewed test of the rule on Indian equities.

This paper asks one question: *had an investor followed the Triple Supertrend on Nifty 50 stocks, would they have ended up with more money than an investor who simply bought and held the same stocks?*

We make three contributions. First, the test is free of survivorship bias: each day's portfolio contains only that day's Nifty 50 members, including companies that later merged, collapsed or were delisted, such as HDFC Ltd, Jaiprakash Associates, Reliance Capital and Ranbaxy. Second, we build a clean, reproducible price panel from NSE's public bhavcopy and corporate-action records rather than a commercial feed, and show that it matches Yahoo Finance's adjusted prices to a median difference of 0.00% a year. Third, parameters are selected on 2010–2018 only and evaluated unchanged on 2019–2026, with autocorrelation-robust and block-bootstrap inference.

The answer is negative for returns and positive for risk. The strategy roughly halves volatility and cuts the worst drawdown by two thirds, but gives up about seven percentage points of annual return. Its risk-adjusted performance is statistically indistinguishable from buy-and-hold.

2 Related literature

The evidence on simple trading rules is mixed and weakens once data snooping is accounted for. Brock et al. (1992) found that moving-average and trading-range rules predicted Dow Jones returns over 1897–1986. Sullivan et al. (1999) showed that much of this apparent predictability disappears once the full universe of rules a researcher could have tried is taken into account. The survey by Park and Irwin (2007) reports that later studies find weaker profits, especially after transaction costs.

Trend following has firmer support at the asset-class level. Moskowitz et al. (2012) document time-series momentum across 58 futures and forward contracts, with a payoff that does well in large market declines. Our results fit that picture: the main benefit of the Supertrend overlay is drawdown control, not extra return.

On method, we follow Bailey and López de Prado (2014) in treating any Sharpe ratio chosen from several tried variants as inflated: we make a single in-sample selection and report every variant we tried. Standard errors follow Newey and West (1987) and the bootstrap uses circular blocks (Politis and Romano, 1992). The Average True Range is due to Wilder (1978); the Supertrend band is commonly attributed to Olivier Seban.

3 Data

Index membership. Point-in-time Nifty 50 membership comes from niftyhistory.in and records 35 changes from April 2010 to September 2026; membership before the first change is inferred by reversing it. The file is internally consistent (no stock is listed both as excluded

and as a member in the same record) and its latest list matches NSE's official constituent list. Over the sample, 98 distinct stocks were members.

Prices. Daily open, high, low and close prices come from NSE's equity bhavcopy files for 4,360 trading days from 2 January 2009 to 9 October 2026. Unlike common free feeds, these files retain companies that were later delisted. Renamed companies (for example Zomato to Eternal) are linked through the issuer code of the ISIN; rights-entitlement lines, which share that code, are excluded.

Corporate actions. Prices are back-adjusted using NSE's corporate-action records (37,010 records, 2009–2026), as summarised in Table 1. After adjustment, the total-return series match Yahoo Finance's adjusted prices to a median difference of 0.00% a year across stocks covered by both. Every remaining one-day move above 20% during index membership was checked by hand and corresponds to a real event, such as March 2020, the Adani sell-off of February 2023 and the collapse of Yes Bank.

Table 1: Price adjustments applied to Nifty 50 member stocks

Event	Method	Events
Split or bonus	Ratio stated in the NSE record; bonus a:b gives a factor $b/(a+b)$	101
Rights issue	Theoretical ex-rights price	17
Demerger	Ex-date open divided by previous close, applied if it moves more than 2%	16
Dividend	Earlier prices scaled by $(1 - D/P)$; dividends reinvested	1,989

Coverage and gaps. Prices exist for 99.3% of the 206,369 stock-days on which a stock was an index member. IDFC Ltd (2010–2015) is recorded under IDFC First Bank's symbol, a different company, and has no usable prices (0.6% of member-days). The Nifty 50 price index (from Yahoo Finance) excludes dividends and is shown for reference only.

4 Methodology

Indicator. For period n and multiplier m, the basic bands are

$$UB_t = \frac{H_t + L_t}{2} + m \cdot ATR_t(n), \qquad LB_t = \frac{H_t + L_t}{2} - m \cdot ATR_t(n), \tag{1}$$

where ATR uses Wilder's smoothing. The final lower band can only rise during an uptrend and the final upper band can only fall during a downtrend. The trend turns up when the close exceeds the previous final upper band and down when it falls below the previous final lower band.

Rule. Each stock is traded independently and long only. The strategy enters when Supertrend $(12, 3)$, $(10, 1)$ and $(11, 2)$ all indicate an uptrend at the close, and exits when the chosen exit Supertrend turns down at the close. Both are executed at the next session's open, so no signal

uses information from the trading day. Each side costs 0.1% of value, covering brokerage, securities transaction tax (STT), stamp duty and slippage. Cash earns nothing while the strategy is out of the market.

Portfolio. With M_t the set of index members with a price on day t and $r_{i,t}$ the strategy (or buy-and-hold) return of stock i, the portfolio return is

$$R_t = \frac{1}{|M_t|} \sum_{i \in M_t} r_{i,t}. \tag{2}$$

The portfolio is rebalanced daily without charge; a stock leaving the index stops contributing from that day.

Design. Six variants (exit Supertrend $(10, 1)$, $(11, 2)$ or $(12, 3)$, each with and without a 200-day moving-average entry filter) are ranked by in-sample (2010–2018) portfolio Sharpe ratio. The best variant is run unchanged on January 2019 to October 2026.

Inference. The mean daily excess return (strategy minus buy-and-hold) is tested with a Newey–West t-statistic with 10 lags. The Sharpe-ratio difference is tested with a circular block bootstrap (20-day blocks, 2,000 resamples, two-sided). Sharpe ratios are annualised with 252 days and a zero risk-free rate.

5 Results

Table 2 shows the in-sample selection. The slowest exit, $(12, 3)$, has the highest Sharpe ratio; the fastest, $(10, 1)$, loses money after costs because it trades on every small pullback. The 200-day filter lowers the Sharpe ratio of every exit.

Table 2: In-sample variant selection, 2010–2018

Exit Supertrend	200-day filter	CAGR	Sharpe	Max drawdown
(12, 3)	No	**2.2%**	**0.30**	**−19.4%**
(11, 2)	No	1.6%	0.27	−16.0%
(12, 3)	Yes	1.6%	0.27	−16.0%
(11, 2)	Yes	1.1%	0.22	−14.9%
(10, 1)	No	−0.1%	0.00	−13.3%
(10, 1)	Yes	−0.3%	−0.07	−13.5%

Table 3 reports the chosen variant in and out of sample, and Figure 1 plots growth and drawdowns. Out of sample, INR 1 invested in the strategy grows to INR 1.80, against INR 2.92 for buy-and-hold. The strategy is invested roughly half the time, so its volatility is about half of buy-and-hold's, as is its return. In March 2020 buy-and-hold fell 38.6% from its high while the strategy, already largely out of the market, fell 11%.

Table 4 tests whether the differences are statistically meaningful. The return shortfall is borderline significant in sample ($t = -2.00$) and not significant out of sample. The Sharpe-ratio

Table 3: Performance of the chosen variant (exit $(12, 3)$), total return

Period	Portfolio	CAGR	Volatility	Sharpe	Max drawdown
2019–2026 (out of sample)	Triple Supertrend	8.0%	8.5%	0.96	−12.1%
	Buy and hold	15.1%	17.4%	0.90	−38.6%
2010–2018 (in sample)	Triple Supertrend	2.2%	8.4%	0.30	−19.4%
	Buy and hold	8.7%	16.5%	0.59	−28.5%

difference is not significant in either period, so we cannot reject that the two approaches have equal risk-adjusted returns.

Table 4: Strategy minus buy-and-hold: significance tests

Period	Mean daily excess	Newey–West t	Sharpe difference [95% CI]	Bootstrap p
2010–2018	−2.8 bp	−2.00	−0.29 [−0.65, 0.06]	0.097
2019–2026	−3.0 bp	−1.58	+0.06 [−0.52, 0.69]	0.996

6 Survivorship bias

Back-filling today's Nifty 50 list into the past roughly doubles buy-and-hold's apparent in-sample return, from 8.7% to 18.6% a year (Figure 2). Today's list consists, by construction, of companies that grew enough to enter or remain in the index; testing on it rewards whichever approach held them longest, which is buy-and-hold. A trend rule that steps aside from falling stocks gains relatively more from including the failures. Backtests of Indian trading rules on today's constituents therefore overstate buy-and-hold by 7–10 percentage points a year and understate the relative case for a trend filter.

7 Robustness and extensions

Data source. Table 5 repeats the test with three price sets. Each gives roughly half the return with a similar Sharpe ratio. With Yahoo prices, which lack seven delisted or merged members (4.7% of member-days), the in-sample choice moves from exit $(12, 3)$ to $(11, 2)$, so the parameter choice is sensitive to data quality.

Table 5: Out-of-sample results (2019–2026) by price source

Prices	Chosen exit	Strategy CAGR	B&H CAGR	Strategy Sharpe	B&H Sharpe
NSE, total return (main)	(12, 3)	8.0%	15.1%	0.96	0.90
NSE, price only	(12, 3)	6.7%	13.1%	0.81	0.79
Yahoo, adjusted	(11, 2)	6.1%	14.8%	0.88	0.88

Exploratory variants. Before the formal design, we compared common variants informally on 2018–2026 daily data for two indices, four Nifty stocks and two cryptocurrencies, with no holdout period. A patient $(12, 3)$ exit raised returns on seven of eight assets; a 200-day filter

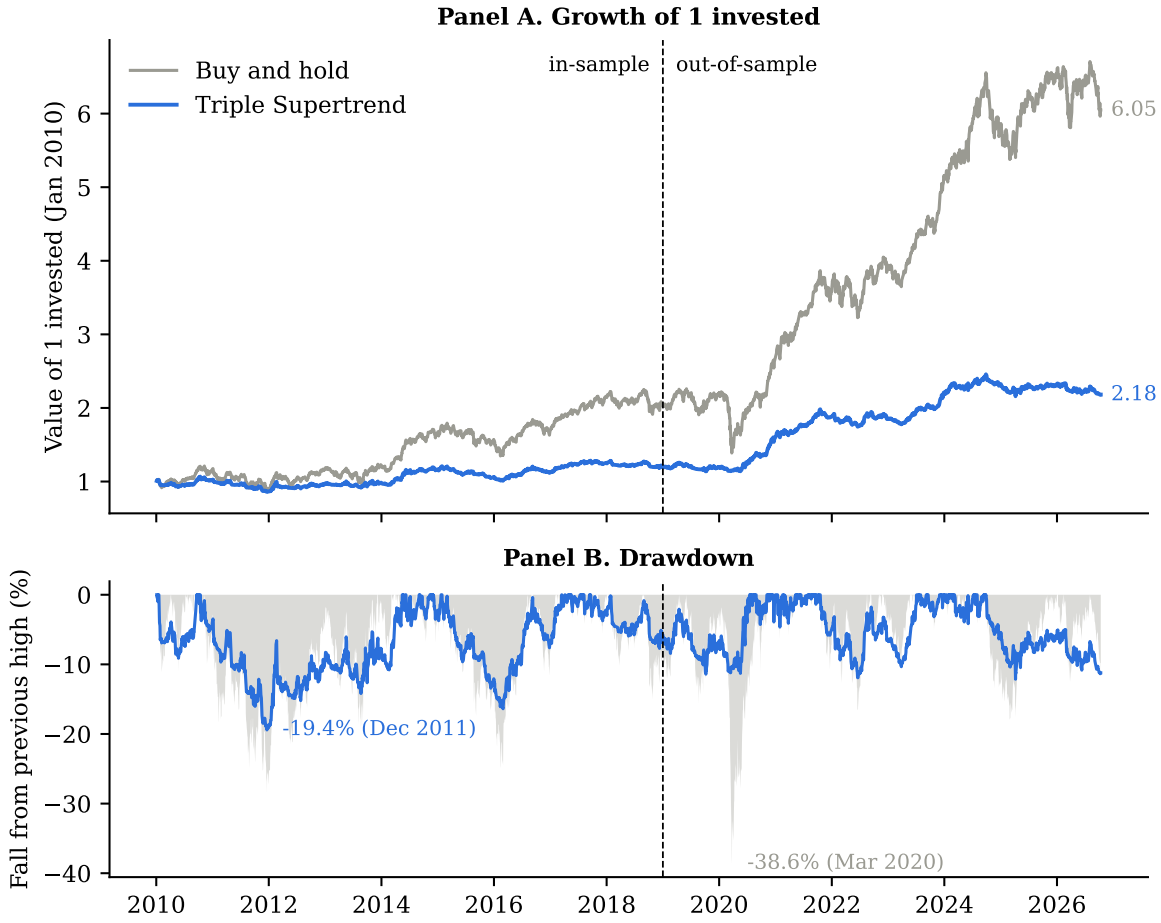


Figure 1: Growth of 1 invested and drawdown from the previous high, equal-weighted point-in-time Nifty 50 members, dividends reinvested, 0.1% cost per trade. The dashed line separates the in-sample (2010–2018) and out-of-sample (2019–2026) periods.

lowered returns on all eight while reducing drawdowns on seven; weekly bars were worse on six of eight; a MACD$(12, 26, 9)$ crossover was mixed and traded two to three times as often; requiring MACD confirmation changed almost nothing; and adding short positions when all three Supertrends turned down hurt most assets. The rule beat buy-and-hold only on Ethereum and, narrowly, Bitcoin.

Index options. Buying at-the-money calls on all-green signals and puts on all-red signals lost money on every index over ten years (Figure 3), with options priced by Black–Scholes from India VIX, one lot per trade and 1% slippage per side. Most contracts expired before the slow exit fired, so each roll paid time value again. Only Nifty calls with at least 45 days to expiry broke even, with a peak-to-trough loss of INR 3.9 lakh along the way.

8 Limitations

The two largest open items both favour the strategy, so the results are, if anything, conservative for it. Cash earns zero while the strategy is out of the market, whereas Indian Treasury bills paid

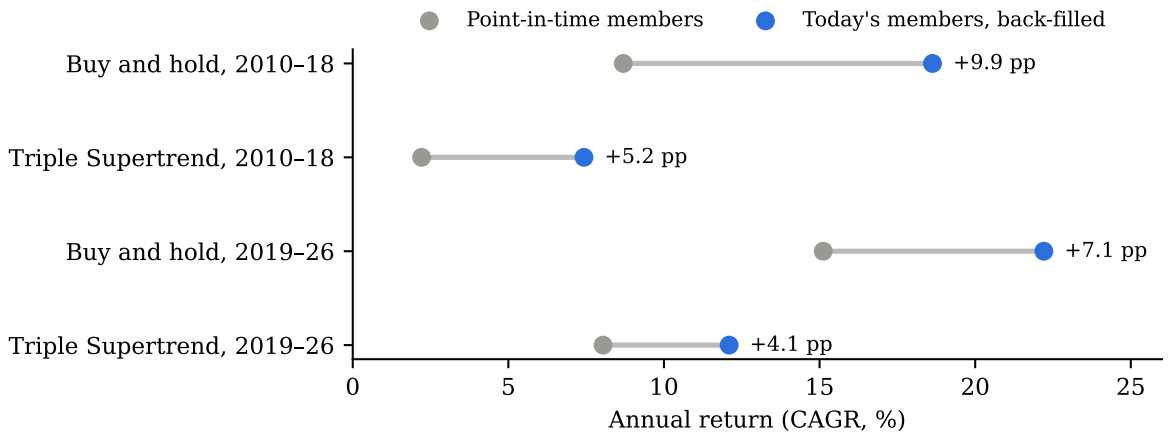


Figure 2: Annual return using point-in-time members versus today's members back-filled into the past.

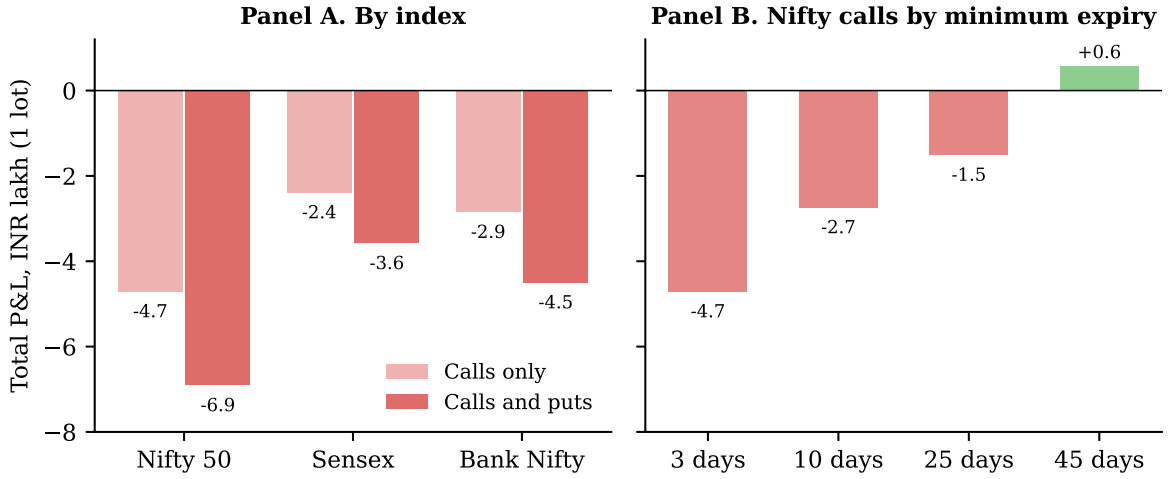


Figure 3: Simulated total profit or loss from buying index options on Triple Supertrend signals over ten years, INR lakh per lot. Model prices, not traded prices.

roughly 4–7% a year; this likely understates the strategy by 2–3 percentage points a year. Nor do we yet compare against a static equity–cash blend of equal volatility, which is the natural benchmark for a rule invested about half the time. Other limitations are a price-only index benchmark, a flat cost model that ignores rebalancing costs, the restriction to the Nifty 50, the missing IDFC Ltd series, rights adjustments that omit face value, 120 percentage-of-face-value dividend records that were skipped, and model rather than traded option prices.

Membership files for the Nifty 100, 200 and 500 from the same source were inconsistent: stocks recorded as removed remained listed in 30 of 46, 42 of 54 and 55 of 75 records respectively, and replaying their change records does not reconcile. Results on those indices are therefore not reported.

9 Conclusion

On survivorship-free, fully adjusted Nifty 50 data, the Triple Supertrend does not beat buy-and-hold. It earns about half the return for about half the risk, and its Sharpe ratio is statistically indistinguishable from buy-and-hold's in both the selection and holdout periods. What the rule reliably delivers is drawdown control: out of sample its worst drawdown is -12% against -39%. Whether it improves on simply holding less equity is the main open question. Two findings matter beyond this rule: backtests on today's Indian index constituents overstate buy-and-hold by 7–10 percentage points a year, and NSE's public files can produce a research-quality panel that includes delisted companies.

Data and code availability

All results can be regenerated with the project's scripts: `fetch_bhavcopy.py` and `fetch_corp_actions.py` download NSE data, `build_nse_series.py` links renames and applies corporate actions and dividends, and `run_pit.py --index nifty50 --source nse_tr` runs the point-in-time test. Figures are produced by `paper/make_figures.py`.

References

Bailey, D. H. and López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting and non-normality. *Journal of Portfolio Management*, 40(5):94–107.

Brock, W., Lakonishok, J., and LeBaron, B. (1992). Simple technical trading rules and the stochastic properties of stock returns. *Journal of Finance*, 47(5):1731–1764.

Moskowitz, T. J., Ooi, Y. H., and Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2):228–250.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Park, C.-H. and Irwin, S. H. (2007). What do we know about the profitability of technical analysis? *Journal of Economic Surveys*, 21(4):786–826.

Politis, D. N. and Romano, J. P. (1992). A circular block-resampling procedure for stationary data. In LePage, R. and Billard, L., editors, *Exploring the Limits of Bootstrap*, pages 263–270. Wiley.

Sullivan, R., Timmermann, A., and White, H. (1999). Data-snooping, technical trading rule performance, and the bootstrap. *Journal of Finance*, 54(5):1647–1691.

Wilder, J. W. (1978). *New Concepts in Technical Trading Systems*. Trend Research.